

Software Requirements Specification

Smart Hostel Management System

Field	Details
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1. Introduction

1.1 Purpose

This document describes, in detail, what the **Smart Hostel Management System** is expected to do. It takes the ideas listed in the project synopsis and turns them into specific, checkable requirements.

The purpose of writing this document is:

- to fix the scope of the project before coding begins, so the work does not keep growing;
- to give the project guide a clear statement of what will be delivered;
- to give the developer a checklist that can be used to confirm the project is complete;
- to serve as the reference document during viva and evaluation.

Anything not written in this document is **not** part of the project.

1.2 Document Conventions

- Requirements are numbered so they can be referred to easily. For example, **FR-5.2** means "Functional Requirement, module 5, point 2".
- The word **shall** means the requirement is compulsory.
- The word **may** means the requirement is optional and can be dropped if time is short.
- Prefixes used in this document:

Prefix	Meaning
FR	Functional Requirement
NFR	Non-Functional Requirement
CON	Design Constraint
ASM	Assumption
TC	Test Case

1.3 Intended Audience

Reader	What they should read
Project Guide / Evaluator	Sections 1, 2, 5, 7, 10
Student / Developer	The whole document, especially 3, 5, 6, 9
External Examiner (viva)	Sections 2.5, 2.6, 2.7, 7, 11

1.4 Project Scope

The Smart Hostel Management System is a **front-end-only website** for a college or university hostel. It brings together, on one website, the information and forms that a hostel resident normally has to look for on a notice board, in the warden's office, or by asking someone.

The website will contain ten sections: Home, Student Registration, Mess Menu, Notice Board, Complaint Portal, Visitor Request, Room Details, Emergency Contacts, Hostel Gallery, and Feedback.

The website is built with **HTML5 and CSS3 only**. There is no server, no database, and no login system. It is a working front-end prototype — the pages, layout, navigation, forms, and validation are real and usable, but the information shown is fixed inside the HTML files, and information typed into forms is not stored anywhere.

This is a deliberate decision, not a shortcoming. The project demonstrates page structure, semantic HTML, form design, layout with Flexbox and Grid, responsive design, and visual design. Storage and processing belong to a later phase, listed in Section 11.

1.5 Definitions, Acronyms and Abbreviations

Term	Meaning
Front-end	The part of a website that runs in the user's browser and that the user can see
Static website	A website whose content is fixed in its files and does not change on its own
Semantic HTML	Using HTML tags that describe meaning (<code>header</code> , <code>nav</code> , <code>main</code> , <code>footer</code>) instead of using <code>div</code> everywhere
Responsive design	Layout that adjusts itself to phone, tablet, and desktop screen sizes
Native validation	Checking done by the browser itself using HTML attributes, without JavaScript

Prototype	A working model built to show how the final system will look and behave
Warden	The staff member in charge of the hostel

1.6 References

1. Project Synopsis — *Smart Hostel Management System*, Shrishti Singh (22515000138)
2. IEEE Std 830-1998, *Recommended Practice for Software Requirements Specifications* (structure adapted for a student project)
3. W3C HTML5 Specification — form controls and native validation
4. W3C CSS3 Specification — Flexbox, Grid, and Media Queries
5. WCAG 2.1 — Web Content Accessibility Guidelines (colour contrast and alternative text)

2. Overall Description

2.1 Product Perspective

This is a new, self-contained product. It does not replace or connect to any existing software in the hostel.

At present, most hostels handle these tasks on paper:

Task today	Problem
Notices pinned on a board	Students who miss the board miss the notice
Mess menu written on a chart	Not visible from the room
Complaints told verbally or written in a register	No record, no acknowledgement
Visitor entry noted in a register	Slow, and no advance information
Emergency numbers on a paper slip	Lost exactly when needed

The website collects all of this into one place that a student can open on a phone.

The system runs entirely inside a web browser. It has no other software dependency — no web server, no runtime, no database engine. Opening `index.html` is enough to run it.

2.2 Product Functions

At a high level, the system shall:

1. Present hostel information (notices, mess menu, room details, emergency contacts, photographs) in a readable, organised form.
2. Provide forms for the four student actions: registration, complaint, visitor request, and feedback.
3. Check the information typed into those forms before allowing submission.
4. Show a clear confirmation after a form is submitted.
5. Work correctly on both a mobile phone and a laptop screen.

2.3 User Classes and Characteristics

User class	Description	Technical skill	How they use the site
Hostel Student (primary)	A resident of the hostel	Basic — can use a phone browser	Reads notices and menu, submits complaints, feedback, and visitor requests
Hostel Warden / Staff (secondary)	Manages the hostel	Basic	Reads the site to see what information students are shown
New / Prospective Student	Has been allotted a hostel seat	Basic	Uses the registration form and gallery
Evaluator / Guide	Reviews the project	High	Inspects pages, code, and this document

All users are treated equally by the system. Because there is no login, there are no roles or permissions in this version.

2.4 Operating Environment

Item	Requirement
Device	Any desktop, laptop, tablet, or smartphone
Operating system	Any (Windows, macOS, Linux, Android, iOS)
Browser	Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari — current version
Internet	Not required. The site runs from local files
Screen width supported	320 px and above
Server software	None
Database	None

2.5 Design and Implementation Constraints

ID	Constraint

CON-1	Only HTML5 and CSS3 shall be used. No JavaScript, no framework, no CSS library such as Bootstrap or Tailwind
CON-2	No backend, no server-side language, and no database
CON-3	All content shown on the site is written directly into the HTML files
CON-4	Form data cannot be stored, sent, emailed, or retrieved
CON-5	The site shall work when opened directly from the file system (<code>file://</code>), not only from a web server
CON-6	No external fonts, icons, or images loaded from the internet — everything shall be kept inside the project folder
CON-7	Every page shall be a separate <code>.html</code> file with its own filename
CON-8	Total project size shall stay small enough to submit easily (target: under 25 MB, mostly images)

2.6 Assumptions and Dependencies

ID	Assumption
ASM-1	The user's browser supports HTML5 form validation and CSS3 Flexbox, Grid, and media queries
ASM-2	Sample data (notices, menu, room list, contacts) is fictional and used only for demonstration
ASM-3	Photographs used in the gallery are either taken by the student, or are royalty-free images used with permission
ASM-4	Real phone numbers of staff are not published; sample numbers are used
ASM-5	The user's browser has JavaScript enabled or disabled — either way the site works, because it does not use JavaScript

2.7 What the System Deliberately Does Not Do

This section exists so that no reviewer expects behaviour that was never promised. The system does **not**:

- store any information typed by the user;
- send an email, SMS, or any message to the warden;
- allow a student or admin to log in;
- track the status of a complaint after submission;
- allocate rooms or update occupancy;
- accept payment of any kind;
- update the mess menu or notices by itself — these are edited by hand in the HTML file.

Each of these appears in Section 11 as a future enhancement.

3. System Structure

3.1 Site Map

```
Home (index.html)
├── Student Registration (register.html)
├── Mess Menu (mess.html)
├── Notice Board (notices.html)
├── Complaint Portal (complaint.html)
├── Visitor Request (visitor.html)
├── Room Details (rooms.html)
├── Emergency Contacts (contacts.html)
├── Hostel Gallery (gallery.html)
├── Feedback (feedback.html)
└── Confirmation (thankyou.html)
    └── ← reached after submitting any of the four forms
```

Eleven pages in total: ten content pages plus one shared confirmation page.

3.2 Folder Structure

Simple, obvious names have been used so that anyone opening the folder can understand it immediately.

```
smart-hostel/
├── index.html
├── register.html
├── mess.html
├── notices.html
├── complaint.html
├── visitor.html
├── rooms.html
├── contacts.html
├── gallery.html
├── feedback.html
├── thankyou.html
├── css/
│   ├── style.css      (layout, colours, header, footer – used by every page)
│   └── forms.css      (styling for all form pages)
├── images/
│   ├── logo.png
│   ├── banner.jpg
│   └── gallery/        (hostel photographs)
└── README.txt         (how to open and run the project)
```

Rule: every page links to `css/style.css`. Form pages additionally link to `css/forms.css`. No page contains a `<style>` block or an inline `style` attribute.

3.3 Common Page Layout

Every page shall follow the same structure so the site feels like one product:

HEADER – logo, hostel name
NAVIGATION – links to all ten pages
PAGE TITLE
MAIN CONTENT (changes per page)
FOOTER – address, contact, copyright line

In HTML terms: `<header>`, `<nav>`, `<main>`, `<footer>`.

3.4 Navigation Rules

ID	Requirement
NAV-1	The navigation bar shall appear on all eleven pages and shall contain the same links in the same order
NAV-2	The link for the page currently open shall be shown differently (different colour or an underline) so the user knows where they are

NAV-3	Clicking the logo or hostel name in the header shall return the user to the Home page
NAV-4	On screens narrower than 768 px the navigation shall stack vertically or wrap, and shall remain fully usable without horizontal scrolling
NAV-5	No page shall be a dead end — every page shall have a working link back to Home

4. External Interface Requirements

4.1 User Interface Requirements

ID	Requirement
UI-1	A single colour scheme shall be used across all pages. Suggested: one main colour, one accent colour, white or off-white background, dark grey text
UI-2	A maximum of two font families shall be used — one for headings, one for body text
UI-3	Body text shall be at least 16 px so it is readable on a phone
UI-4	Every clickable item shall show a visible change on hover (desktop) and on focus (keyboard)
UI-5	Form fields shall have a visible label above or beside them — placeholder text alone is not acceptable as a label
UI-6	Error and success messages produced by the browser shall not be hidden by CSS
UI-7	Page content shall be centred with a maximum width (around 1200 px) so lines of text do not stretch across a wide monitor
UI-8	Spacing shall be consistent — the same gap between sections on every page

4.2 Hardware Interfaces

The system does not communicate with any hardware. It requires only a display, and a mouse or touchscreen.

4.3 Software Interfaces

The only software interface is the web browser's HTML and CSS rendering engine. The system does not call any API, library, or external service.

4.4 Communication Interfaces

None. The site does not use a network connection at any point. The only exception is the `tel:` links in Emergency Contacts, which hand the number over to the device's own phone application.

5. Functional Requirements

Each module below lists what the page must contain and how it must behave.

FR-1 — Home Page (`index.html`)

Purpose: Introduce the hostel and let the user reach any other section quickly.

ID	Requirement
FR-1.1	The page shall display the hostel name and a banner image or coloured banner section
FR-1.2	The page shall display a short welcome paragraph of 3–5 lines describing the hostel
FR-1.3	The page shall display an overview section with at least four facts about the hostel — for example: number of blocks, total rooms, mess timings, and warden's name
FR-1.4	The page shall display quick-link cards to at least six sections: Notices, Mess Menu, Complaint, Visitor Request, Room Details, Emergency Contacts
FR-1.5	Each quick-link card shall be fully clickable and shall open the correct page
FR-1.6	The page shall display hostel rules or timings in a short list (in time, out time, mess timings, visitor hours)
FR-1.7	Quick-link cards shall arrange themselves in three columns on desktop, two on tablet, and one on mobile

FR-2 — Student Registration (`register.html`)

Purpose: Collect the details of a student joining the hostel.

ID	Requirement
FR-2.1	The page shall present a form containing every field listed in Section 6.1
FR-2.2	Fields marked compulsory in Section 6.1 shall use the <code>required</code> attribute so the browser blocks submission when they are empty
FR-2.3	The email field shall use <code>type="email"</code> and the phone fields shall use <code>type="tel"</code> with a 10-digit <code>pattern</code>
FR-2.4	The form shall be divided into three visible groups with sub-headings: Personal Details , Academic Details , Guardian Details
FR-2.5	The form shall include a Reset button that clears all fields
FR-2.6	The form shall include a Submit button which, when all validation passes, opens <code>thankyou.html</code>
FR-2.7	A visible note shall appear near the form stating that this is a demonstration form and no data is saved
FR-2.8	On mobile, fields shall occupy the full width of the screen and shall stack in a single column

FR-3 — Mess Menu (`mess.html`)

Purpose: Show the weekly food schedule.

ID	Requirement
FR-3.1	The page shall display a table with seven rows, one for each day of the week
FR-3.2	The table shall have four content columns: Breakfast, Lunch, Snacks, Dinner
FR-3.3	The table shall have a clearly styled header row that is visually separated from the body
FR-3.4	Alternate rows shall have a different background colour so the table is easy to read across
FR-3.5	One row may be highlighted as "today". This highlight is applied by hand in the HTML — the page cannot detect the real day, since that would need JavaScript
FR-3.6	The page shall display mess timings for all four meals below the table
FR-3.7	The page shall display a notes area for special items or holiday menus
FR-3.8	On screens narrower than 600 px the table shall become horizontally scrollable rather than shrinking to unreadable text

FR-4 — Notice Board (`notices.html`)

Purpose: Show hostel announcements.

ID	Requirement
FR-4.1	The page shall display at least six sample notices

FR-4.2	Each notice shall show: date, title, body text, and the person or office that issued it
FR-4.3	Notices shall be arranged newest first, from top to bottom
FR-4.4	Each notice shall carry a category tag — for example <i>Urgent, Mess, Maintenance, Event, General</i>
FR-4.5	Each category tag shall have its own colour, and a legend explaining the colours shall appear at the top of the page
FR-4.6	Notices tagged <i>Urgent</i> shall be visually distinct — for example, a coloured left border
FR-4.7	Each notice shall be presented as a separate card with clear spacing between cards

FR-5 — Complaint Portal (`complaint.html`)

Purpose: Let a student report a maintenance problem.

ID	Requirement
FR-5.1	The page shall present a form containing every field listed in Section 6.2
FR-5.2	The form shall include a category dropdown with at least these options: Electrical, Plumbing, Furniture, Cleanliness, Internet, Water Supply, Other
FR-5.3	The form shall include a priority choice using radio buttons: Low, Medium, High
FR-5.4	The form shall include a description textarea with a minimum length of 20 characters, enforced using <code>minLength</code>
FR-5.5	The description field shall be at least four rows tall
FR-5.6	Compulsory fields shall block submission when empty
FR-5.7	A successful submission shall open <code>thankyou.html</code>
FR-5.8	The page shall display a short "How complaints are handled" section listing the expected steps and response time
FR-5.9	A visible note shall state that the complaint is not actually recorded in this version

FR-6 — Visitor Request (`visitor.html`)

Purpose: Let a student request permission for a visitor.

ID	Requirement
FR-6.1	The page shall present a form containing every field listed in Section 6.3
FR-6.2	The visit date field shall use <code>type="date"</code> and the time fields shall use <code>type="time"</code>
FR-6.3	The form shall include a relationship dropdown: Parent, Guardian, Sibling, Relative, Friend, Other
FR-6.4	The form shall collect both the student's details and the visitor's details, under two separate sub-headings
FR-6.5	The page shall display the hostel's visitor rules — visiting hours, ID requirement, and maximum number of visitors
FR-6.6	A successful submission shall open <code>thankyou.html</code>
FR-6.7	A visible note shall state that submitting this form does not grant permission, and that approval is a future feature

FR-7 — Room Details (`rooms.html`)

Purpose: Show what rooms exist and which are free.

ID	Requirement
FR-7.1	The page shall display a table of at least twelve rooms
FR-7.2	Each room row shall show: Room Number, Block, Floor, Room Type, Capacity, Currently Occupied, Status
FR-7.3	Room Type shall be one of: Single, Double, Triple
FR-7.4	Status shall be one of: Available, Partially Filled, Full
FR-7.5	Each status shall have its own background colour, and a legend explaining the colours shall appear above the table
FR-7.6	The page shall display a summary line above the table — total rooms, total capacity, and number of available rooms
FR-7.7	Rooms shall be grouped or sorted by block, and then by room number
FR-7.8	On small screens the table shall be horizontally scrollable

FR-8 — Emergency Contacts (`contacts.html`)

Purpose: Put important numbers where they can be found quickly.

ID	Requirement
FR-8.1	The page shall display contact cards for at least these roles: Warden, Assistant Warden, Security Gate, Ambulance / Medical Room, Electrician, Plumber, Fire Emergency, Police
FR-8.2	Each card shall show the role, the person's name where applicable, the phone number, and availability hours
FR-8.3	Each phone number shall be a clickable <code>tel:</code> link so that a phone user can dial directly
FR-8.4	Contacts available around the clock shall be marked "24 × 7"
FR-8.5	The most critical contacts (Ambulance, Fire, Security) shall appear first and shall be visually emphasised
FR-8.6	Cards shall arrange themselves in three columns on desktop, two on tablet, one on mobile

FR-9 — Hostel Gallery (`gallery.html`)

Purpose: Show the hostel facilities.

ID	Requirement
FR-9.1	The page shall display at least eight photographs in a grid
FR-9.2	The grid shall use CSS Grid and shall reflow to fewer columns on smaller screens
FR-9.3	Every image shall have a caption describing what it shows — for example <i>Reading Room</i> , <i>Mess Hall</i> , <i>Common Room</i> , <i>Washing Area</i>
FR-9.4	Every image shall have a meaningful <code>alt</code> attribute
FR-9.5	Images shall be of consistent size within the grid, using <code>object-fit: cover</code> so that they are not stretched
FR-9.6	Hovering over an image shall produce a visible effect such as a slight zoom or an overlay
FR-9.7	The page may include a CSS-only enlarged view using the <code>:target</code> selector. This is optional and shall be dropped if it causes layout problems
FR-9.8	All images shall be compressed before use, and no single image shall exceed 500 KB

FR-10 — Feedback (`feedback.html`)

Purpose: Collect suggestions from students.

ID	Requirement
FR-10.1	The page shall present a form containing every field listed in Section 6.4
FR-10.2	The form shall include a rating from 1 to 5 using radio buttons
FR-10.3	The form shall include a feedback category dropdown: Mess Food, Cleanliness, Maintenance, Security, Wi-Fi, Staff Behaviour, Other
FR-10.4	The form shall include a checkbox allowing the student to submit anonymously
FR-10.5	When "submit anonymously" is intended, the name and roll number fields shall not be marked <code>required</code> , so the form can still be submitted without them
FR-10.6	The comments textarea shall allow at least 300 characters
FR-10.7	A successful submission shall open <code>thankyou.html</code>

FR-11 — Confirmation Page (`thankyou.html`)

Purpose: Tell the user their submission was accepted by the form.

ID	Requirement
FR-11.1	The page shall display a clear success heading, for example "Submitted Successfully"
FR-11.2	The page shall display a short message thanking the user
FR-11.3	The page shall clearly state that this is a demonstration and that no data has been stored

FR-11.4	The page shall provide a button linking back to the Home page
FR-11.5	The same confirmation page shall be shared by all four forms

6. Content and Field Definitions

This section defines exactly what each form contains. It is the reference used while writing the HTML.

6.1 Student Registration Fields

Field	Input type	Compulsory	Rule
Full Name	text	Yes	3-50 characters
Roll Number	text	Yes	Exactly 11 digits
Date of Birth	date	Yes	—
Gender	radio	Yes	Male / Female / Other
Email	email	Yes	Must contain @ and a domain
Mobile Number	tel	Yes	Exactly 10 digits
Course	select	Yes	B.Tech / BCA / B.Sc / MBA / Other
Branch	text	Yes	—
Year of Study	select	Yes	1 / 2 / 3 / 4
Home Address	textarea	Yes	Minimum 10 characters
City	text	Yes	—
State	text	Yes	—
Pin Code	text	Yes	Exactly 6 digits
Guardian Name	text	Yes	—
Guardian Mobile	tel	Yes	Exactly 10 digits
Room Preference	select	No	Single / Double / Triple / No preference
Blood Group	select	No	A+, A-, B+, B-, O+, O-, AB+, AB-

6.2 Complaint Form Fields

Field	Input type	Compulsory	Rule
Student Name	text	Yes	—
Roll Number	text	Yes	Exactly 11 digits
Room Number	text	Yes	—
Block	select	Yes	A / B / C / D
Mobile Number	tel	Yes	Exactly 10 digits
Complaint Category	select	Yes	Per FR-5.2
Priority	radio	Yes	Low / Medium / High
Date of Issue	date	Yes	—
Description	textarea	Yes	Minimum 20 characters

6.3 Visitor Request Fields

Field	Input type	Compulsory	Rule
Student Name	text	Yes	—
Roll Number	text	Yes	Exactly 11 digits
Room Number	text	Yes	—
Student Mobile	tel	Yes	Exactly 10 digits
Visitor Name	text	Yes	—
Visitor Mobile	tel	Yes	Exactly 10 digits
Relationship	select	Yes	Per FR-6.3

Visitor ID Type	select	Yes	Aadhaar / Voter ID / Driving Licence / Other
Date of Visit	date	Yes	—
Expected Arrival Time	time	Yes	—
Expected Departure Time	time	Yes	—
Number of Visitors	number	Yes	Between 1 and 5
Purpose of Visit	textarea	Yes	Minimum 10 characters

6.4 Feedback Form Fields

Field	Input type	Compulsory	Rule
Student Name	text	No	See FR-10.5
Roll Number	text	No	See FR-10.5
Feedback Category	select	Yes	Per FR-10.3
Overall Rating	radio	Yes	1 to 5
Comments / Suggestions	textarea	Yes	Minimum 10 characters, maximum 300
Submit Anonymously	checkbox	No	—

6.5 Fixed Content Written into the Pages

Page	Content to be written by hand
Mess Menu	7 days × 4 meals = 28 entries, plus timings
Notice Board	At least 6 notices with date, title, body, issuer, category
Room Details	At least 12 rooms with all seven columns filled
Emergency Contacts	At least 8 contacts with role, name, number, availability
Gallery	At least 8 images with captions and alt text
Home	Overview facts, rules, timings

7. Non-Functional Requirements

7.1 Usability

ID	Requirement
NFR-1.1	Any section of the site shall be reachable from any page in a single click
NFR-1.2	A first-time user shall be able to find the mess menu without instructions
NFR-1.3	Every page shall have a heading that states clearly what the page is
NFR-1.4	Form errors shall be visible without the user having to scroll
NFR-1.5	No page shall require horizontal scrolling on a 320 px wide screen, except tables covered by FR-3.8 and FR-7.8

7.2 Performance

ID	Requirement
NFR-2.1	Each page shall render in under one second when opened from local files on a normal laptop
NFR-2.2	The total size of all CSS files shall stay under 100 KB
NFR-2.3	The gallery page shall stay under 5 MB in total image weight

Note: these numbers are achievable because the site is static. No claim is made about behaviour under network load, since there is no network.

7.3 Portability and Responsiveness

ID	Requirement
NFR-3.1	The site shall display correctly in Chrome, Firefox, Edge, and Safari
NFR-3.2	The site shall be usable on screens from 320 px to 1920 px wide
NFR-3.3	At least three breakpoints shall be used: below 600 px (mobile), 600–1024 px (tablet), above 1024 px (desktop)
NFR-3.4	The site shall run without any installation, build step, or server

7.4 Accessibility

ID	Requirement
NFR-4.1	Every image shall have an <code>alt</code> attribute
NFR-4.2	Every form field shall have a <code><label></code> connected to it using the <code>for</code> attribute
NFR-4.3	Text shall have a contrast ratio of at least 4.5 : 1 against its background
NFR-4.4	The whole site shall be navigable using only the Tab key
NFR-4.5	Colour alone shall not be used to convey meaning — status colours shall also carry a text label

7.5 Maintainability

ID	Requirement
NFR-5.1	Colours, fonts, and spacing values shall be defined once as CSS custom properties in <code>:root</code> and reused everywhere
NFR-5.2	Class names shall describe purpose, not appearance — <code>notice-card</code> , not <code>blue-box</code>
NFR-5.3	Each CSS file shall be divided into commented sections
NFR-5.4	The HTML shall be properly indented and shall pass the W3C validator with no errors
NFR-5.5	Changing the main colour of the site shall require editing one line of CSS

7.6 Reliability

ID	Requirement
NFR-6.1	No link on any page shall be broken
NFR-6.2	No image on any page shall be missing

7.7 Security

The system holds no user data, has no login, and makes no network request. Therefore it has **no security requirements**, and no security claim is made in this document.

This is stated openly rather than left unsaid: a static site with unsaved forms cannot leak data, but it also cannot protect anything. Security becomes relevant only when the enhancements in Section 11 are added.

8. Design Constraints and Coding Standards

ID	Standard
STD-1	Every page shall begin with <code><!DOCTYPE html></code> and declare <code>lang="en"</code>
STD-2	Every page shall include <code><meta charset="UTF-8"></code> and the responsive viewport meta tag
STD-3	Every page shall have a unique, descriptive <code><title></code>
STD-4	Semantic tags shall be used: <code>header</code> , <code>nav</code> , <code>main</code> , <code>section</code> , <code>article</code> , <code>footer</code>
STD-5	No inline <code>style</code> attributes and no <code><style></code> blocks inside HTML files
STD-6	No table shall be used for page layout — tables are only for tabular data (Mess Menu, Room Details)
STD-7	File and folder names shall be lowercase with no spaces
STD-8	Layout shall use Flexbox and CSS Grid, not floats
STD-9	Comments shall be added at the top of each HTML file stating which page it is
STD-10	Deprecated tags such as <code><center></code> , <code></code> , and <code><marquee></code> shall not be used

9. Validation Rules

Because JavaScript is not used, all validation is done by the browser using HTML5 attributes. This table lists exactly which attribute produces which check.

Check needed	Attribute used	Example
Field must be filled	<code>required</code>	<code><input type="text" required></code>
Valid email format	<code>type="email"</code>	<code><input type="email" required></code>
10-digit phone number	<code>pattern="[0-9]{10}"</code>	<code><input type="tel" pattern="[0-9]{10}"></code>
11-digit roll number	<code>pattern="[0-9]{11}"</code>	<code><input type="text" pattern="[0-9]{11}"></code>
6-digit pin code	<code>pattern="[0-9]{6}"</code>	<code><input type="text" pattern="[0-9]{6}"></code>
Minimum text length	<code>minlength</code>	<code><textarea minlength="20"></code>
Maximum text length	<code>maxlength</code>	<code><textarea maxlength="300"></code>
Number within a range	<code>min</code> and <code>max</code>	<code><input type="number" min="1" max="5"></code>
Valid date	<code>type="date"</code>	<code><input type="date" required></code>
Valid time	<code>type="time"</code>	<code><input type="time" required></code>
Helpful error message	<code>title</code>	<code>title="Enter a 10-digit mobile number"</code>

Behaviour on submit: if any check fails, the browser blocks submission and shows its own message next to the field. If all checks pass, the browser follows the form's `action` and opens `thankyou.html`.

Limitation to state honestly at viva: these checks confirm the *shape* of the data, not its truth. The form cannot tell whether a roll number actually exists, or whether a date is in the past. Those checks need JavaScript or a backend, and are listed in Section 11.

10. Test Cases and Traceability

10.1 Test Cases

ID	Test	Steps	Expected result	Covers
TC-01	Navigation works everywhere	Open each of the 11 pages, click every nav link	Every link opens the correct page; no 404	NAV-1, NFR-6.1
TC-02	Current page is indicated	Open Mess Menu	The Mess Menu nav link looks different from the others	NAV-2
TC-03	Empty registration blocked	Open register.html, click Submit without typing anything	Browser blocks submission and highlights the first empty required field	FR-2.2
TC-04	Invalid email rejected	Type abc in the email field and submit	Browser shows an email format message	FR-2.3
TC-05	Phone length enforced	Type a 9-digit mobile number and submit	Browser blocks submission	FR-2.3
TC-06	Valid registration accepted	Fill all compulsory fields correctly and submit	thankyou.html opens	FR-2.6, FR-11.5
TC-07	Reset clears the form	Fill several fields, click Reset	All fields become empty	FR-2.5
TC-08	Short complaint blocked	Type a 5-character description and submit	Browser blocks submission due to minlength	FR-5.4
TC-09	Complaint category present	Open the category dropdown	All seven options listed in FR-5.2 are present	FR-5.2
TC-10	Priority is single-choice	Select High, then select Low	Only one option stays selected	FR-5.3
TC-11	Mess table complete	Open mess.html	7 day rows × 4 meal columns are all filled	FR-3.1, FR-3.2
TC-12	Mess table scrolls on mobile	Set browser width to 375 px	Table scrolls sideways; page itself does not	FR-3.8
TC-13	Notices ordered correctly	Open notices.html	Dates run newest at the top to oldest at the bottom	FR-4.3
TC-14	Category colours match legend	Compare each notice tag with the legend	Every tag colour matches its legend entry	FR-4.5
TC-15	Visitor count limited	Enter 7 in Number of Visitors and submit	Browser blocks submission	FR-6.1, Section 6.3
TC-16	Room legend matches table	Open rooms.html	Every status colour in the table appears in the legend	FR-7.5
TC-17	Room summary is correct	Count the rows and compare with the summary line	Counts match	FR-7.6
TC-18	Phone links dial	Open contacts.html on a phone, tap a number	The phone dialler opens with the number filled	FR-8.3
TC-19	Images all load	Open gallery.html	Eight or more images display; no broken image icons	FR-9.1, NFR-6.2
TC-20	Alt text present	View page source of gallery.html	Every has a non-empty alt	FR-9.4, NFR-4.1
TC-21	Anonymous feedback works	Tick anonymous, leave name and roll blank, submit	Form submits successfully	FR-10.5
TC-22	Rating is single-choice	Select 4, then select 2	Only one rating stays selected	FR-10.2
TC-23	Confirmation page states the limitation	Submit any form	thankyou.html clearly says no data was stored	FR-11.3
TC-24	Responsive layout at 375 px	View every page at 375 px width	No horizontal page scrolling; content readable	NFR-3.2, NFR-1.5
TC-25	Responsive layout at 768 px	View every page at 768 px width	Layout switches to the tablet arrangement	NFR-3.3

TC-26	Keyboard navigation	Use only the Tab key on every page	All links and fields can be reached; focus is visible	NFR-4.4, UI-4
TC-27	HTML validity	Run each page through the W3C validator	No errors reported	NFR-5.4
TC-28	Colour change is central	Change one CSS variable in <code>:root</code>	The colour updates across all pages	NFR-5.1, NFR-5.5
TC-29	No inline styles	Search all HTML files for <code>style=</code>	No results found	STD-5
TC-30	Works offline	Disconnect the internet, open index.html	Site loads and functions completely	NFR-3.4, CON-6

10.2 Traceability — Synopsis to Requirements

Objective from the synopsis	Requirements that deliver it
Digitise hostel-related services	FR-1 to FR-10
Reduce manual paperwork	FR-2, FR-5, FR-6, FR-10
Improve communication between students and administration	FR-4, FR-5, FR-8, FR-10
Provide easy access to hostel information	FR-1, FR-3, FR-4, FR-7, FR-8, FR-9, NAV-1
Clean, responsive, user-friendly interface	UI-1 to UI-8, NFR-1, NFR-3, NFR-4

Every module in the synopsis maps to exactly one requirement group:

Synopsis module	Requirement group
Home Page	FR-1
Student Registration	FR-2
Mess Menu	FR-3
Notice Board	FR-4
Complaint Portal	FR-5
Visitor Request	FR-6
Room Details	FR-7
Emergency Contacts	FR-8
Hostel Gallery	FR-9
Feedback	FR-10

11. Future Enhancements

These are **not** part of the current project. They are listed to show the direction of further work.

Phase 2 — Adding JavaScript (still front-end only)

Enhancement	What it would add
Custom form validation	Error messages designed by the developer instead of the browser's default ones
Live search and filter	Searching notices by keyword, filtering rooms by block or availability
Auto-highlight today's meal	The mess table would detect the real day, removing the manual step in FR-3.5
Working image lightbox	A proper gallery viewer with next and previous controls
Local storage	Submitted complaints and feedback would survive a page refresh on the same device
Mobile menu	A hamburger menu with smooth open and close behaviour
Character counter	Live count of characters remaining in the feedback box

Phase 3 — Adding a Backend and Database

Enhancement	What it would add
Student and Admin login	Separate views and permissions for students and the warden
Database integration	Registrations, complaints, visitor requests, and feedback stored permanently
Complaint status tracking	Status moving through Submitted → In Progress → Resolved, visible to the student
Visitor approval system	The warden approves or rejects requests; the student sees the result
Room allocation	Rooms assigned to students; occupancy updated automatically
Attendance management	Daily in and out records
Notice management	The warden posts notices through a form instead of editing HTML
Email and SMS notifications	Automatic alerts for notices and complaint updates
Online fee payment	Payment through a payment gateway with receipts
Inventory management	Records of hostel furniture and equipment
Reports and analytics	Complaint counts by category, feedback averages, occupancy trends

Note: the synopsis listed "Mobile Responsive Design" as a future enhancement. In this SRS it has been moved into the current scope (NFR-3), because the project objectives already promise a responsive interface. This resolves a contradiction in the synopsis.

12. Conclusion

The Smart Hostel Management System, as specified in this document, is a complete front-end website of eleven pages that covers the ten hostel functions identified in the synopsis. It is built entirely with HTML5 and CSS3.

The document sets clear boundaries. It states what the system does, what it does not do, and why. Forms validate input using the browser's own checks and lead to a confirmation page, but store nothing — and this is written openly on the pages themselves rather than being hidden.

Within these boundaries the project is genuinely useful as a demonstration. It shows semantic page structure, form design, table design, responsive layout with Flexbox and Grid, a consistent visual system driven by CSS variables, and attention to accessibility. Thirty test cases are defined, each traceable back to a requirement, and each requirement traceable back to an objective in the synopsis.

The system is a solid foundation. The enhancements in Section 11 describe how it can grow — first by adding JavaScript for interactivity, then by adding a backend and database to make it a full hostel management system.

Appendix A — Glossary

Term	Meaning
Breakpoint	A screen width at which the layout changes
CSS variable	A value defined once and reused throughout the stylesheet
Flexbox	A CSS layout method for arranging items in a row or column
Grid	A CSS layout method for arranging items in rows and columns together
Media query	A CSS rule that applies only at certain screen sizes
object-fit	A CSS property that controls how an image fills its box
Placeholder	Grey hint text inside an empty input field
:root	The top-level CSS selector, used to declare site-wide variables
Semantic tag	An HTML tag whose name describes the meaning of its content
:target	A CSS selector matching the element whose id is in the page URL
W3C Validator	An online tool that checks HTML for errors

Appendix B — Page-wise Content Checklist

Tick each item as the page is completed.

index.html - ☐ Header with logo and hostel name - ☐ Navigation bar with all 10 links - ☐ Banner section - ☐ Welcome paragraph - ☐ Overview facts (4 or more) - ☐ Quick-link cards (6 or more) - ☐ Rules and timings list - ☐ Footer

register.html - ☐ All 17 fields from Section 6.1 - ☐ Three grouped sections with sub-headings - ☐ `required` on all compulsory fields - ☐ Patterns on phone, roll number, pin code - ☐ Submit and Reset buttons - ☐ "Not saved" note

mess.html - ☐ 7 × 4 menu table filled completely - ☐ Styled header row - ☐ Alternating row colours - ☐ Mess timings - ☐ Notes area - ☐ Horizontal scroll on mobile

notices.html - ☐ 6 or more notice cards - ☐ Date, title, body, issuer on each - ☐ Category tags with colours - ☐ Legend at the top - ☐ Newest first

complaint.html - ☐ All 9 fields from Section 6.2 - ☐ Category dropdown with 7 options - ☐ Priority radio buttons - ☐ Description with `minLength="20"` - ☐ "How complaints are handled" section

visitor.html - ☐ All 13 fields from Section 6.3 - ☐ Student and visitor sections separated - ☐ Date and time inputs - ☐ Visitor rules displayed

rooms.html - ☐ 12 or more rooms in the table - ☐ All 7 columns filled - ☐ Status colours plus legend - ☐ Summary line above the table

contacts.html - ☐ 8 or more contact cards - ☐ `tel:` links on every number - ☐ 24 × 7 markings - ☐ Critical contacts placed first

gallery.html - ☐ 8 or more images - ☐ Captions on all - ☐ Alt text on all - ☐ Grid reflows on smaller screens - ☐ Hover effect

feedback.html - ☐ All 6 fields from Section 6.4 - ☐ 1-5 rating radios - ☐ Category dropdown - ☐ Anonymous checkbox - ☐ 300-character limit

thankyou.html - ☐ Success heading - ☐ Thank-you message - ☐ "No data stored" note - ☐ Back to Home button

End of Software Requirements Specification — Smart Hostel Management System, Version 1.0